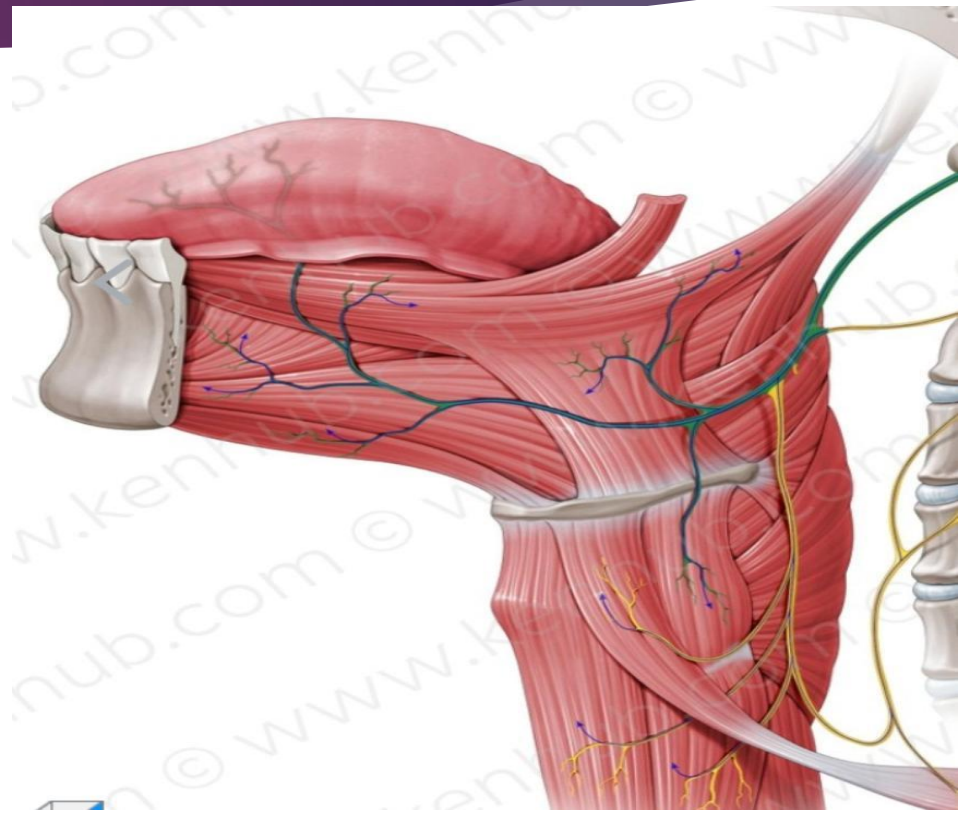
THE TONGUE ANATOMY

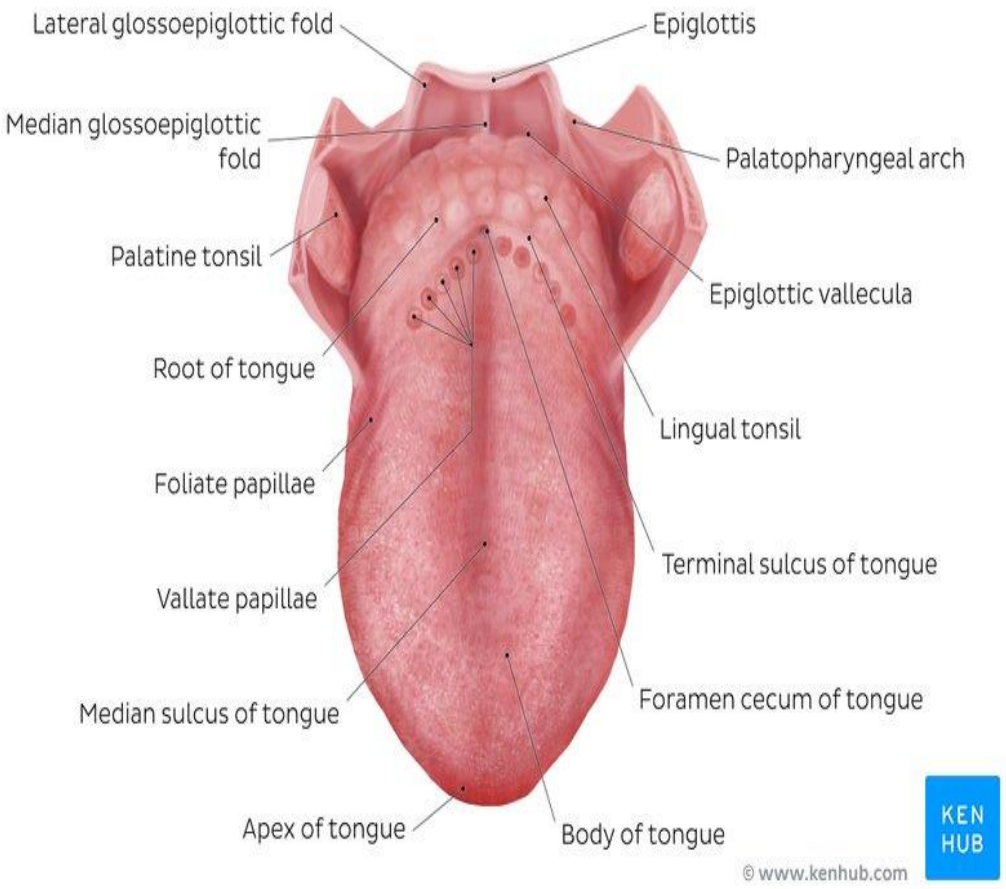
INTRODUCTION

- Tongue is a muscular organ
- Made up of skeletal muscles
- Hence attached to bones like mandible and hyoid
- It is situated in floor of oral cavity
- Tongue is covered with mucous membrane
- It is lined with stratified squamous epithelium
- Keratinised at oral part
- Non keratinised at pharyngeal part



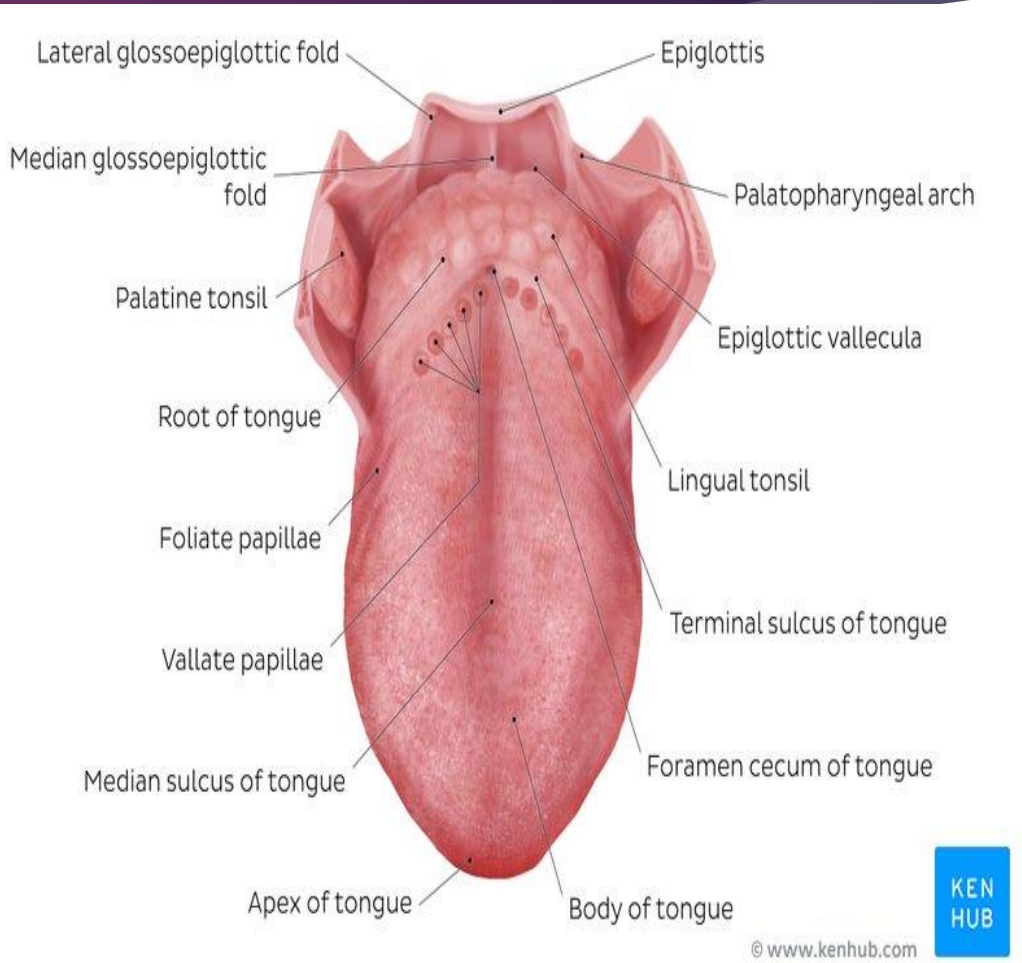
Structures related to the Tongue

- Tongue limited anteriorly and laterally by upper and lower set of teeth
- Superiorly: bordered by hard and soft palate
- Inferiorly: continuous with mucosa of oral cavity
- Palatoglossal arch: it's is a fold of mucous membrane that connect the posterior part of tongue with soft palate , posterior to this fold is tonsilar fossa where the Palatine tonsil located...and posteriorly limited by another fold known as palatopharyngeal Arch



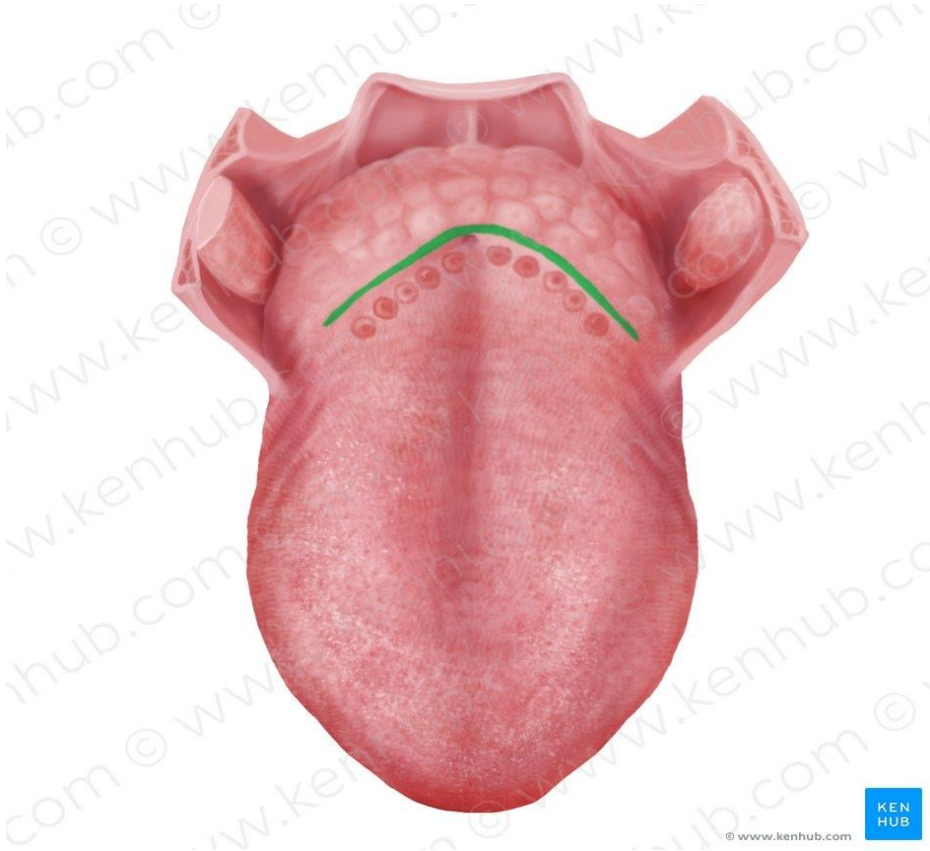
PARTS OF TONGUE

- Three parts :
- Root/Base : Attached end ,it is the most posterior part
- Tip/apex: Free end at lower incisors
- Body :main bulk of muscle tissue of tongue.
- It possess two surfaces
- Dorsum and ventral surface



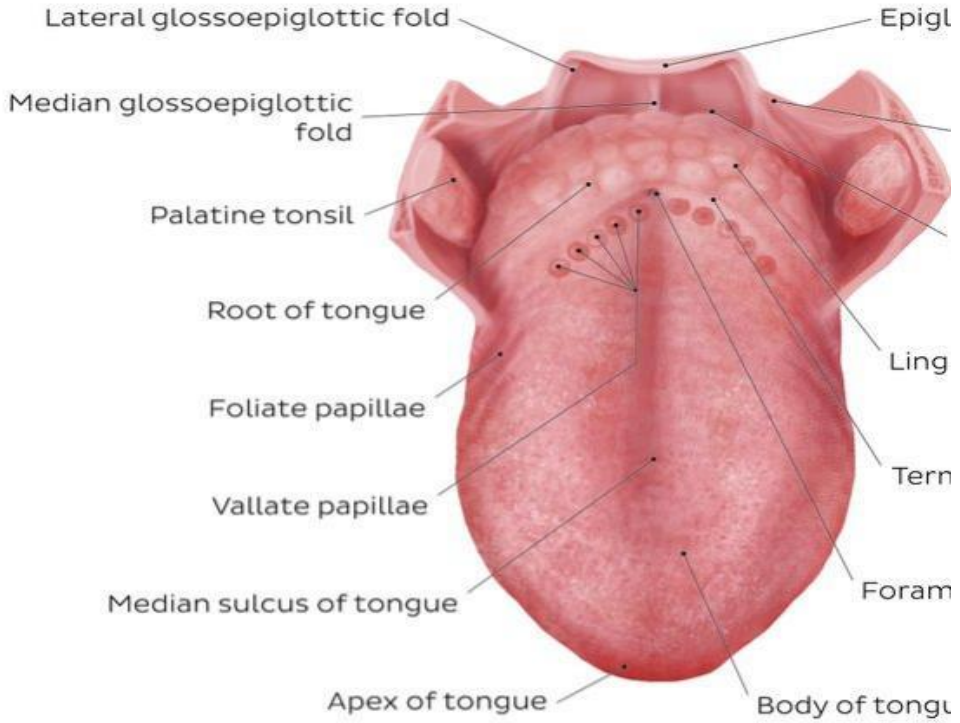
Dorsum

- Shows an inverted v' shaped sulcus _Sulcus Terminalis
- Centre median depressed area of sulcus foramen caecum
- It is the embryological remnant of developing thyroglossal duct from which thyroid gland developes.
- Sulcus divides the dorsum



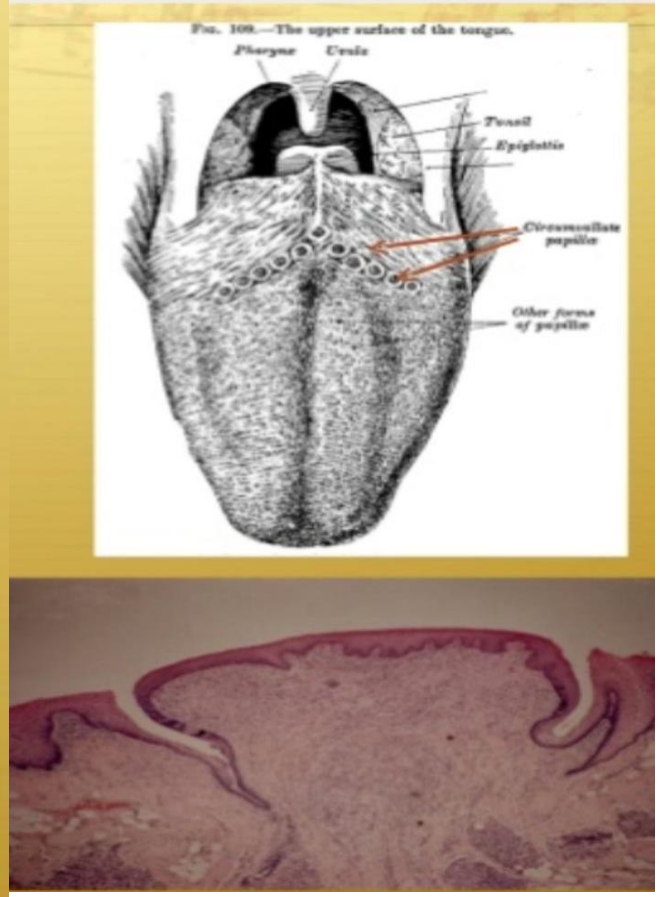
Dorsum

- Papillae: mucous membrane shows epithelial projections
- They give the tongue a velvety appearance
- 3 types of papillae
- Circumvallate
- Fungiform
- Filiform



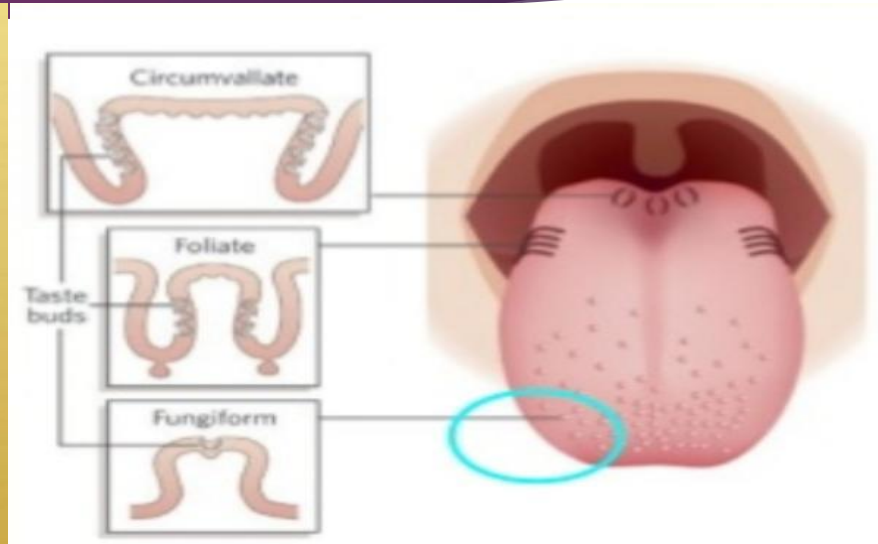
Circumvallate papillae

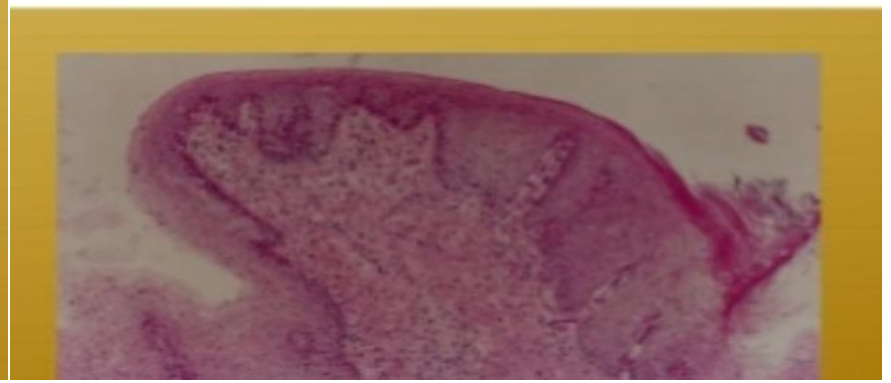
- Large cylindrical structures
- 8 to 12 in number
- Form a ‘V’ shaped row in front of sulcus terminalis on the dorsal surface of the tongue
- The entire structure is covered with squamous epithelium, in both sulcal walls & taste buds around



Fungiform papillae

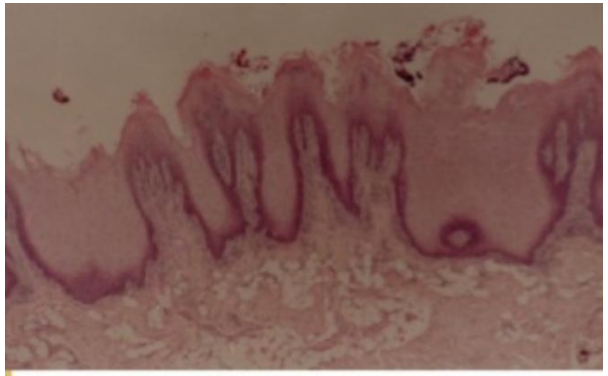
- Located mainly on the lingual margin
- Differ from filiform because are larger, rounded and deep red in colour
- Bears one or more taste buds on its apical surface
- These are mushroom shaped, more numerous near tip & margins of tongue but some of them scattered over the dorsum

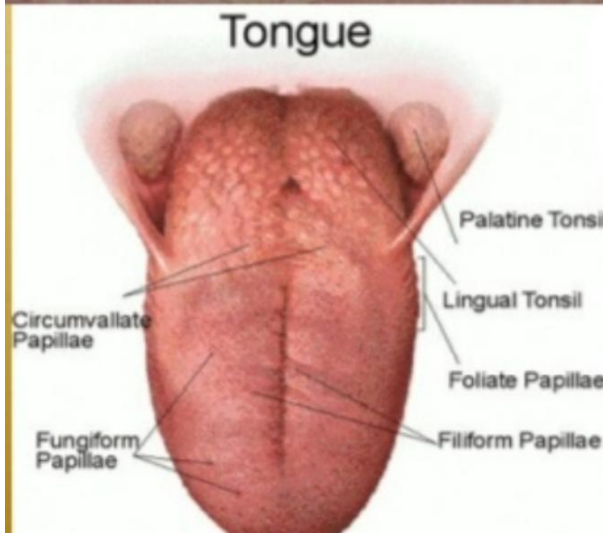




Filiform papillae

- Minute, conical, cylindrical projections which cover most of the presulcul dorsal area.
- Increase the friction between the tongue and food
- They bear many secondary papillae which are more pointed than those of vallate and fungiform papillae and covered with keratin



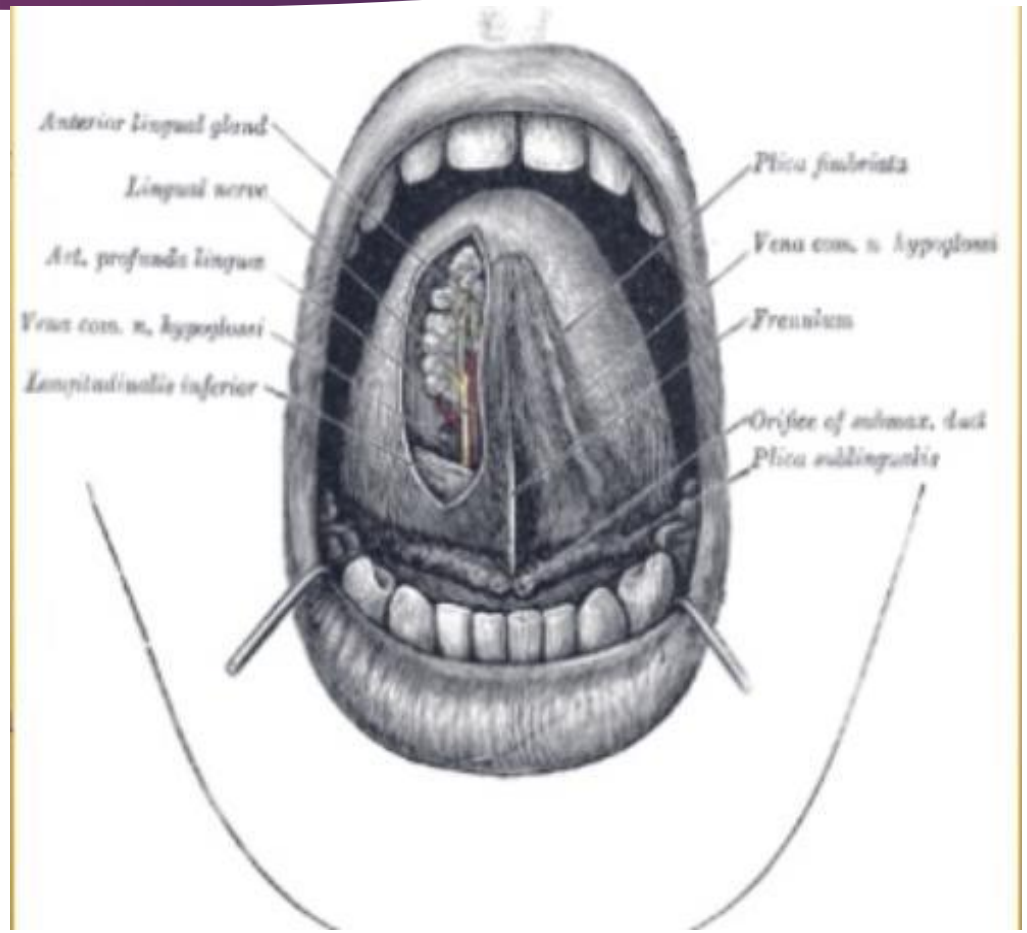


Foliate papillae

- Additionally a fold of mucous membrane present at the lateral margin just anterior to sulcus

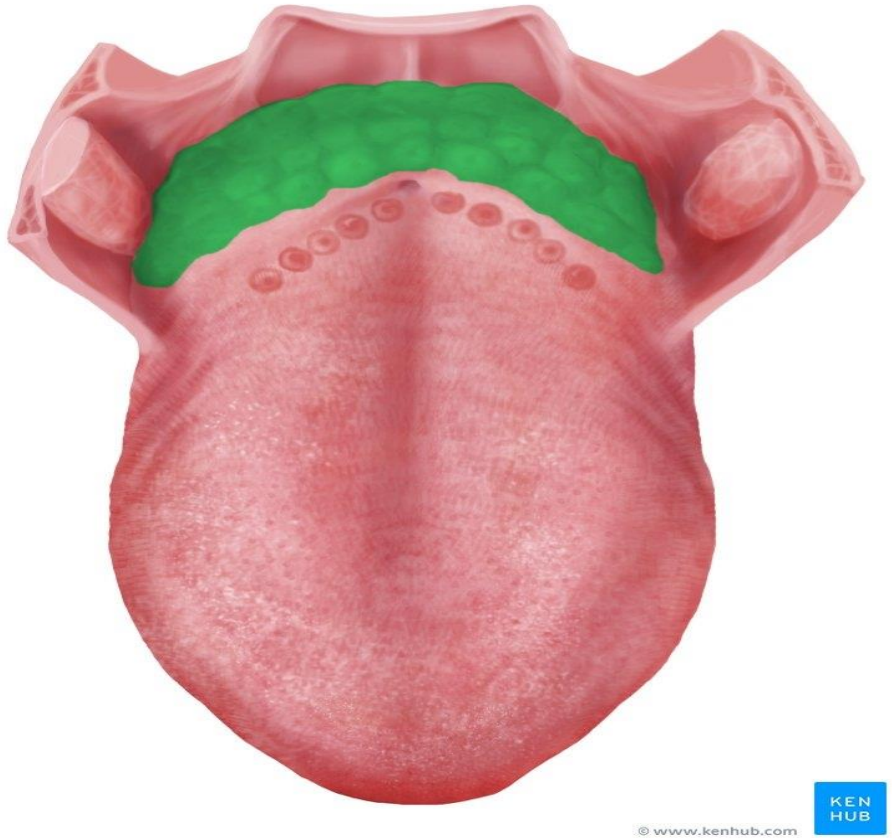
VENTRAL SURFACE

- Presenting a median fold of mucous membrane known as frenulum linguae
- Lateral to it another fold of mucous membrane present on both sides and is known as plica fimbriata which gives a frilled appearance
- On either side and laterally to the frenulum there is deep lingual vein Which shows a bluish coloration.



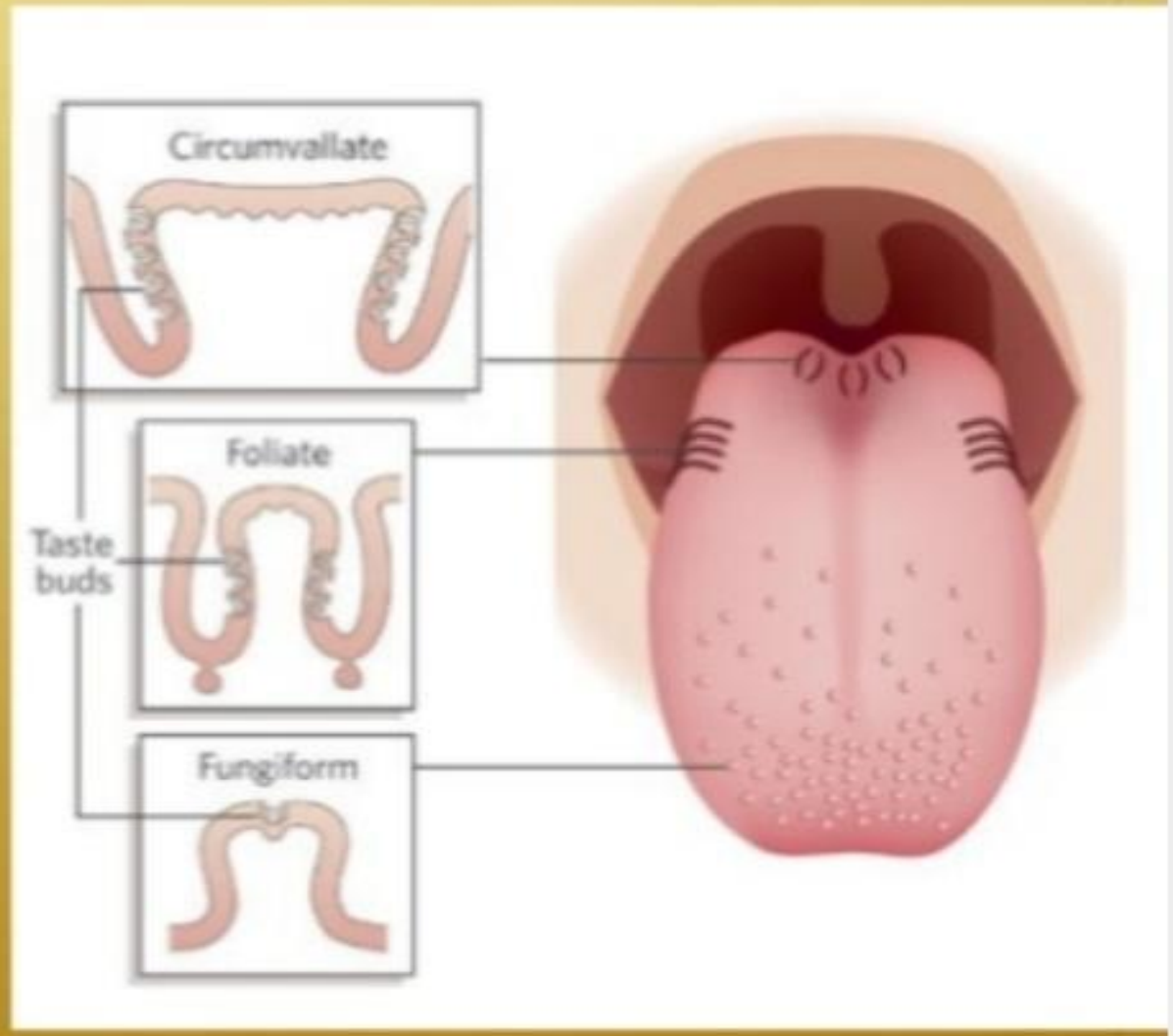
Posterior 1/3of tongue

- is the anterior wall of the oropharynx
- extends from sulcus terminalis and epiglottis
- Foramen caecum: is a small depression at the apex of the sulcus - the remains of the upper end of the thyroglossal duct
  - ✓ there are no papillae behind this sulcus
- Smooth mucous membrane has a nodular appearance - constitute the lingual tonsil, part of Waldeyer's ring.
- Between tongue and epiglottis - midline flange of mucous membrane (median glossoepiglottic fold).
  - ✓ Each side of which is depression (valleculae), bounded laterally by similar mucosal fold (lateral glossoepiglottic fold), extending from the side of the epiglottis to the wall of the pharynx



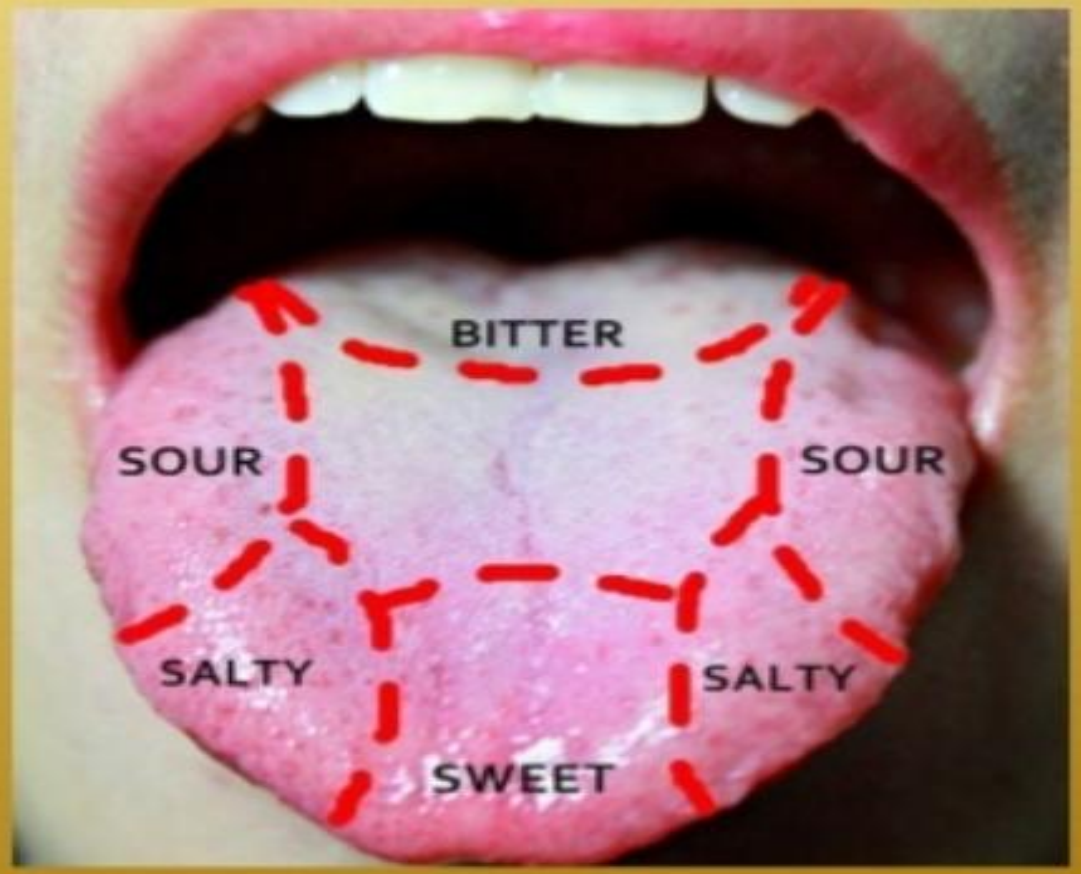
Taste buds

- Present in relation to cirumvallate papilla, fungiform papillae and foliate papilla
- Also present on the soft palate, the epiglottis, the palatoglossal arches, and the posterior wall of the oropharynx



Taste discrimination

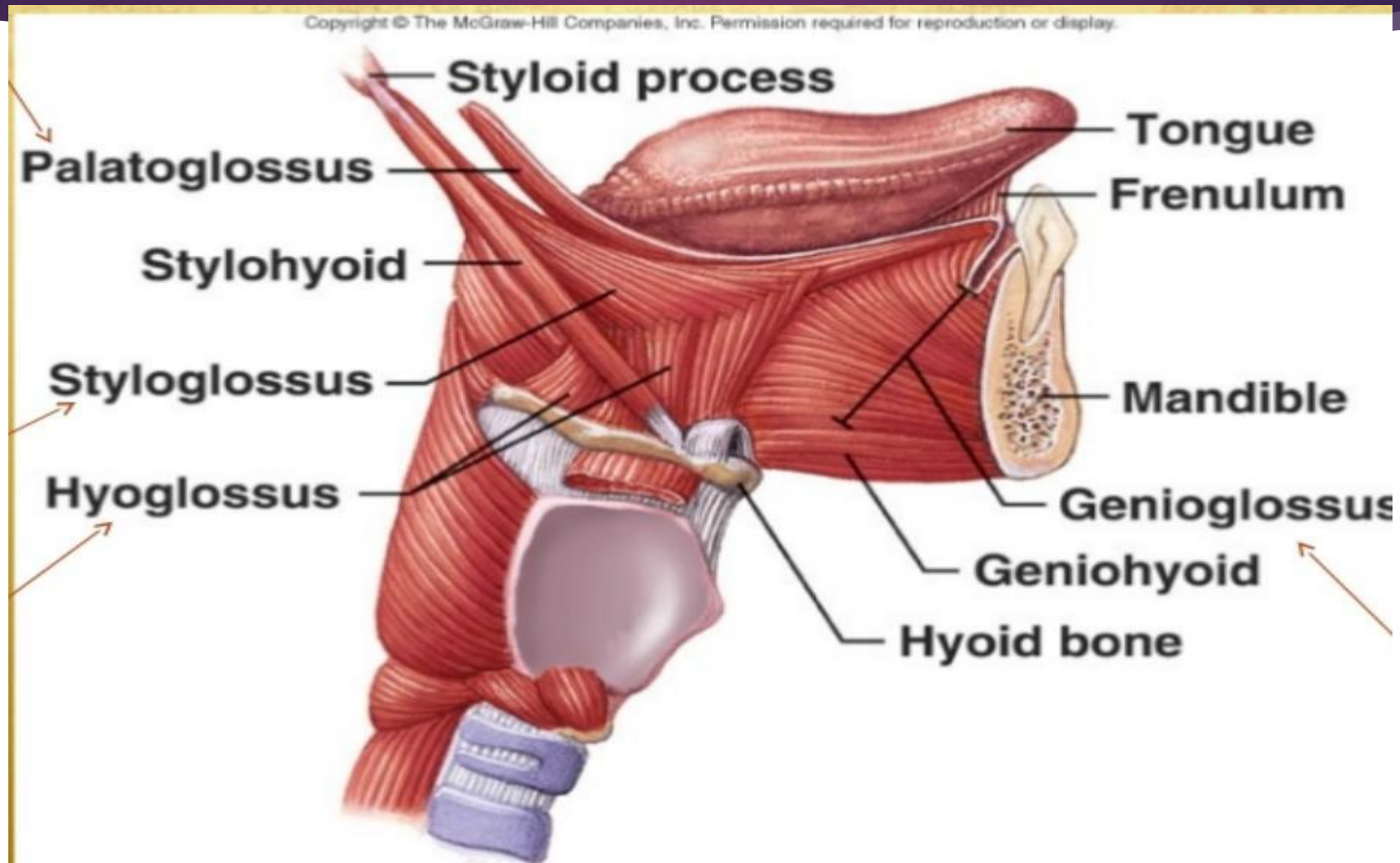
- Gustatory receptors detect four main types of taste sensation
- Sweet: tip
- Sour: middle
- Salty: anterolateral
- Bitter: base
- However recent evidence indicates that all areas of tongue are responsive to all taste stimuli



Muscles of tongue

- Two Types
- Intrinsic
- Muscles that present inside the tongue and regulates the shape
- Extrinsic
- Muscles that anchor tongue to Surrounding bony structures




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Styloid process
Palatoglossus
Stylohyoid
Styloglossus
Hyoglossus
Tongue
Frenulum
Mandible
Genioglossus
Geniohyoid
Hyoid bone

Genioglossus muscle

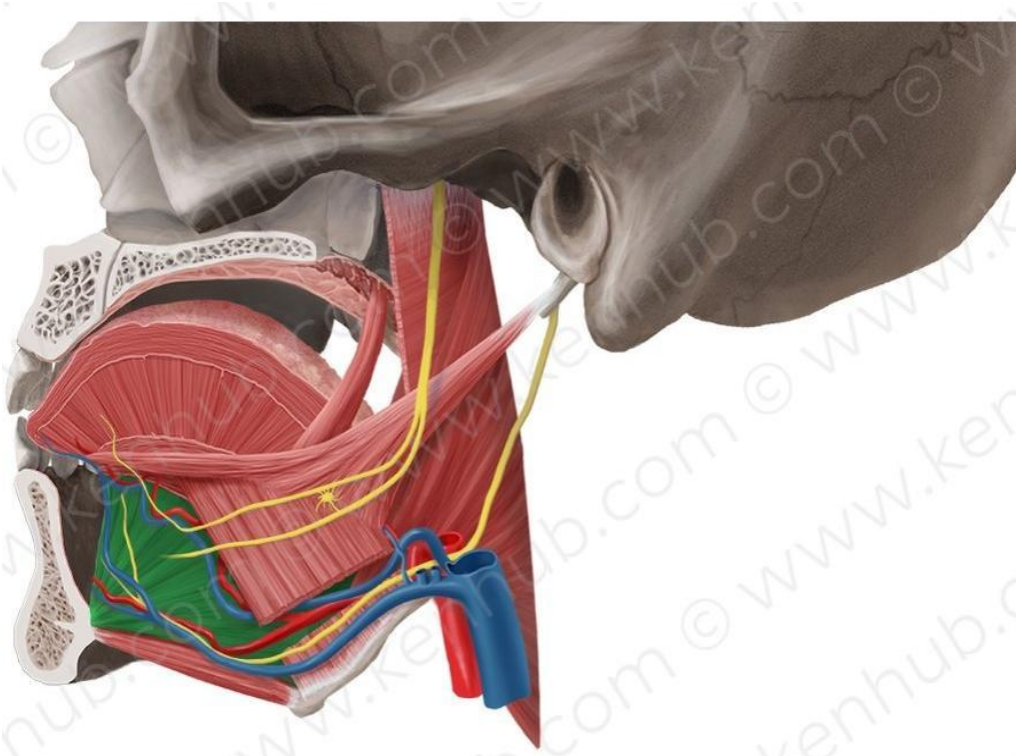
Origin - Superior mental spine of mandible

Insertion - entire length of dorsum of tongue, lingual aponeurosis, body of hyoid bone

Innervation - hypoglossal nerve (CN XII)

Blood supply - sublingual branch of lingual artery, submental branch of facial artery

Action - depresses and protrudes tongue (bilateral contraction); deviates tongue contralaterally (unilateral contraction)



Hyoglossus muscle

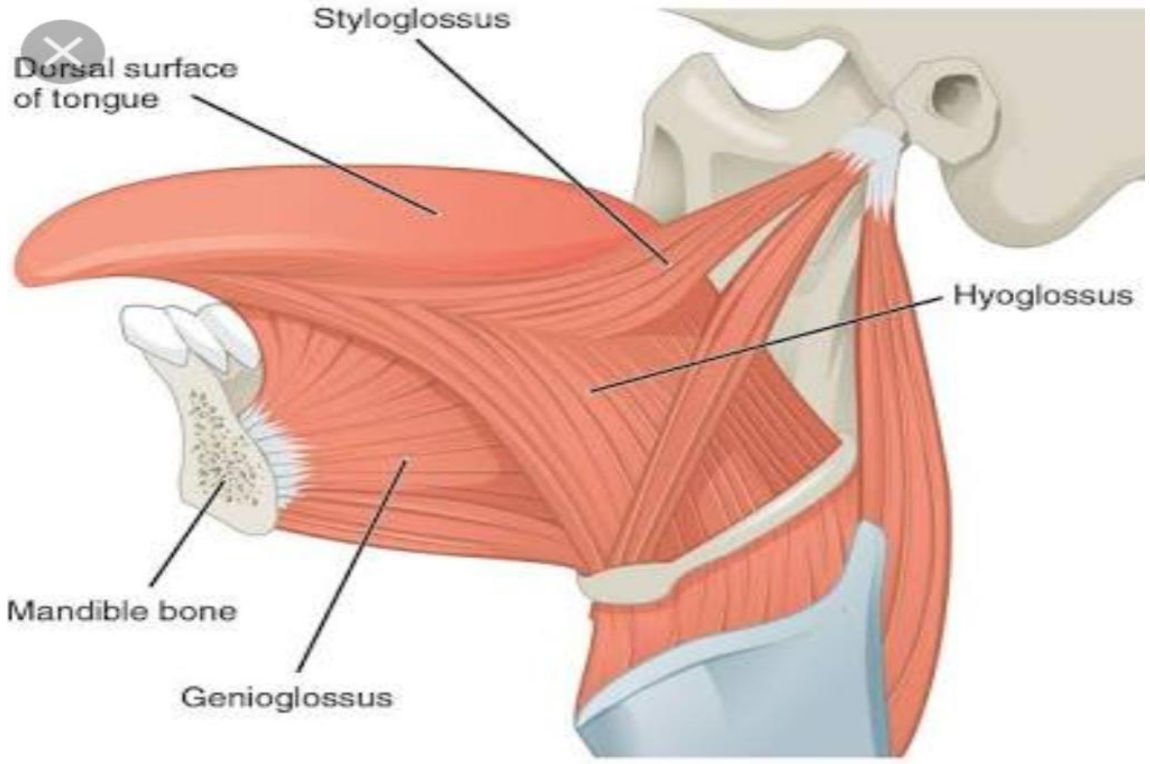
Origin - body and greater horn of hyoid bone

Insertion - inferior/ventral parts of lateral tongue

Innervation - hypoglossal nerve (CN XII)

Blood supply - sublingual branch of lingual artery, submental branch of facial artery

Action - depresses and retracts tongue



Styloglossus muscle

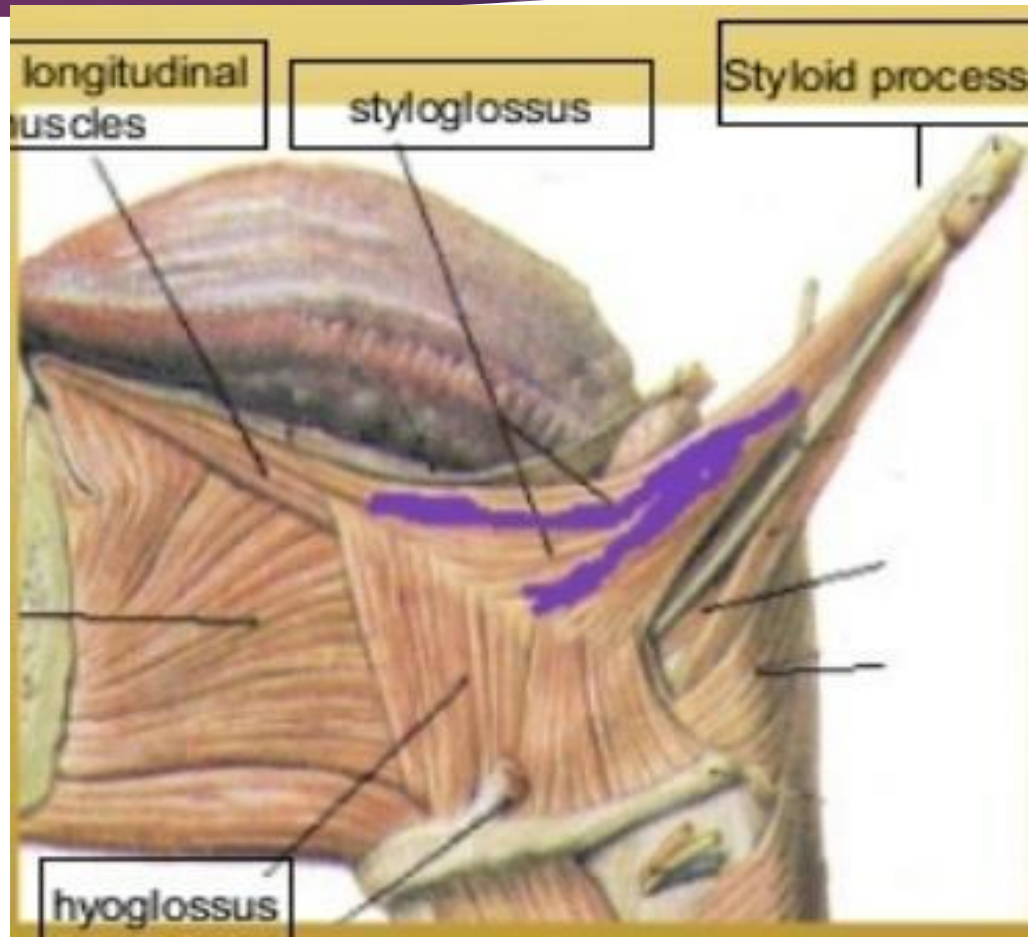
Origin - anterolateral aspect of styloid process (of temporal bone), stylomandibular ligament

Insertion - blends with inferior longitudinal muscle (longitudinal part); blends with hyoglossus muscle (oblique part)

Innervation - hypoglossal nerve (CN XII)

Blood supply - sublingual branch of lingual artery

Action - retracts and elevates lateral aspects of tongue



Palatoglossal muscle

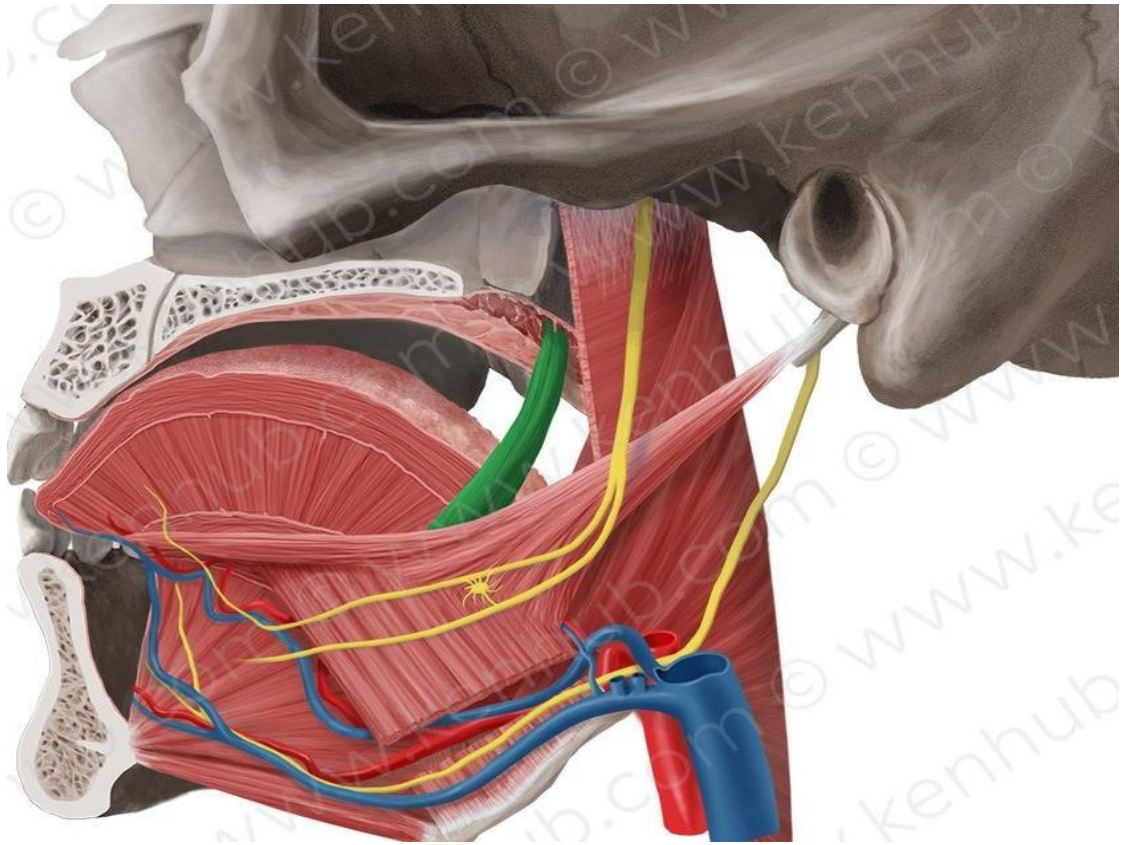
Origin - palatine aponeurosis of soft palate

Insertion - lateral margins of tongue, blends with intrinsic muscles of tongue

Innervation - vagus nerve (CN X) (via branches of pharyngeal plexus)

Blood supply - ascending palatine branch of facial artery, ascending pharyngeal artery

Action - elevates root of tongue, constricts isthmus of fauces



Intrinsic muscles

Superior longitudinal

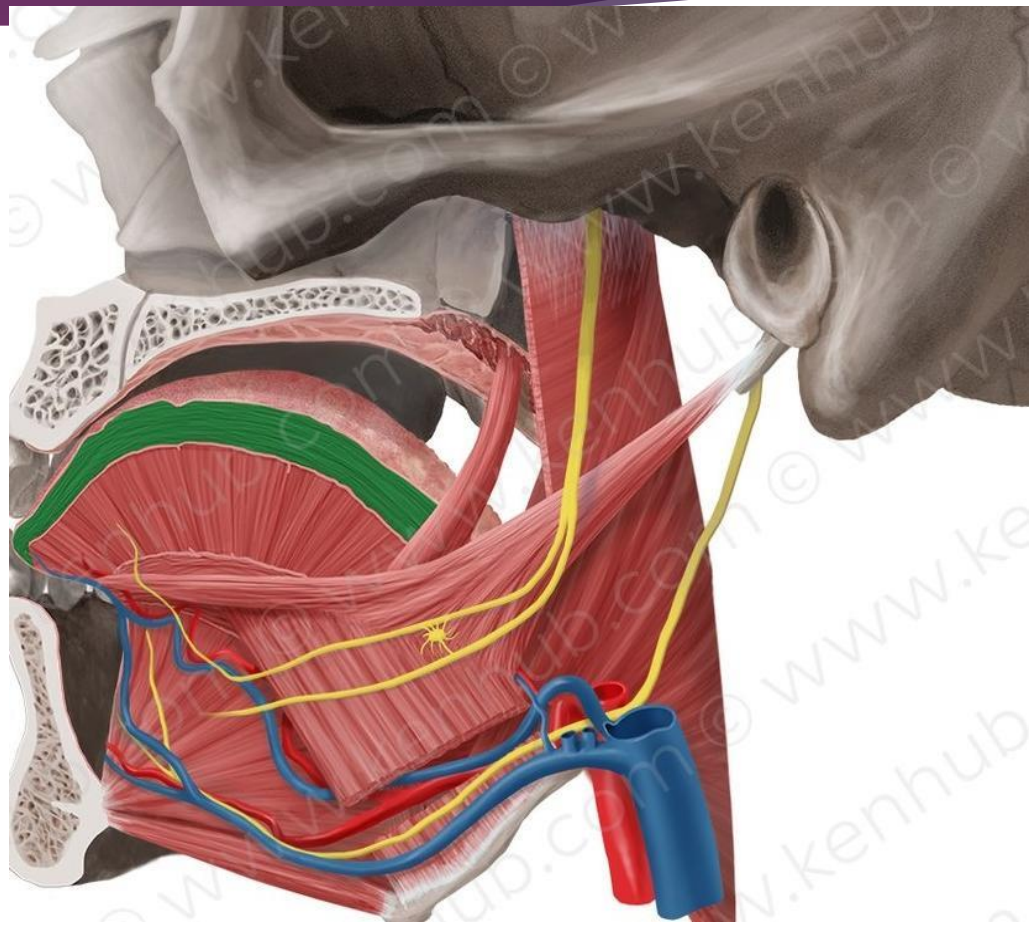
Origin - submucosa of posterior tongue, lingual septum

Insertion - apex/anterolateral margins of tongue

Innervation - hypoglossal nerve (CN XII)

Blood supply - lingual branch of external carotid artery

Action - retracts and broadens tongue, elevates apex of tongue



Inferior longitudinal

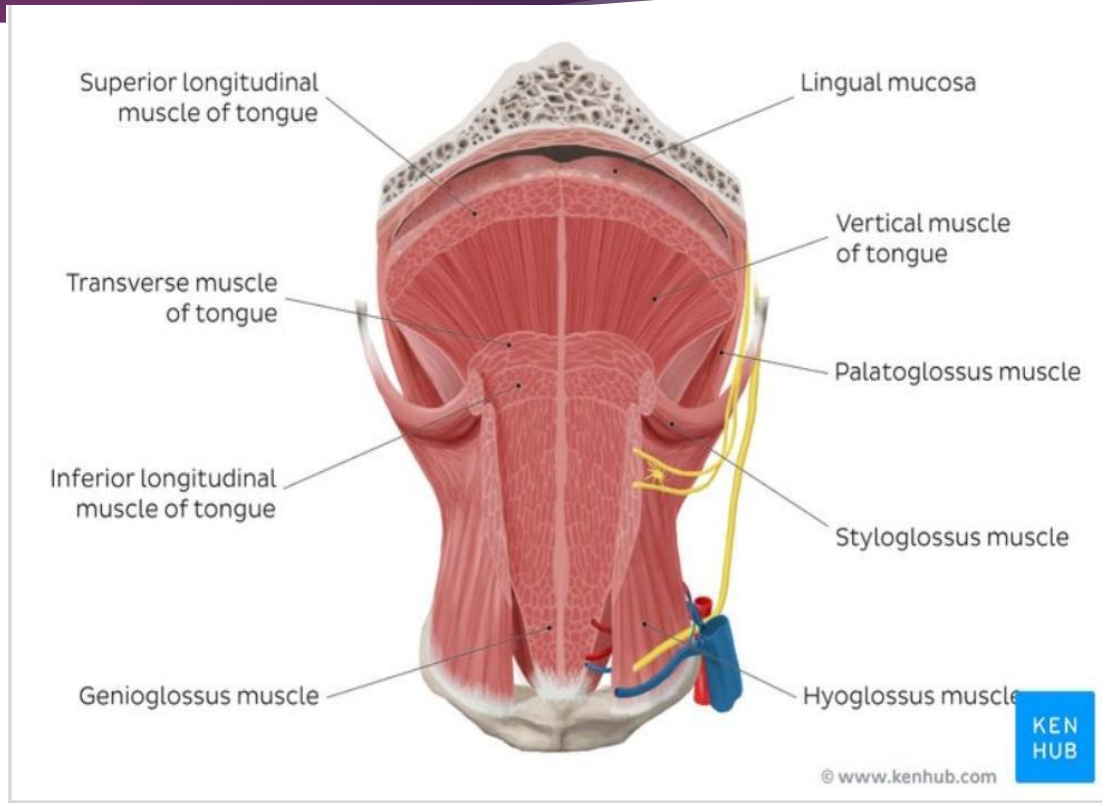
Origin - root of tongue, body of hyoid bone

Insertion - apex of tongue

Innervation - hypoglossal nerve (CN XII)

Blood supply - lingual branch of external carotid artery

Action - retracts and broadens tongue, lowers apex of tongue



Transverse muscle

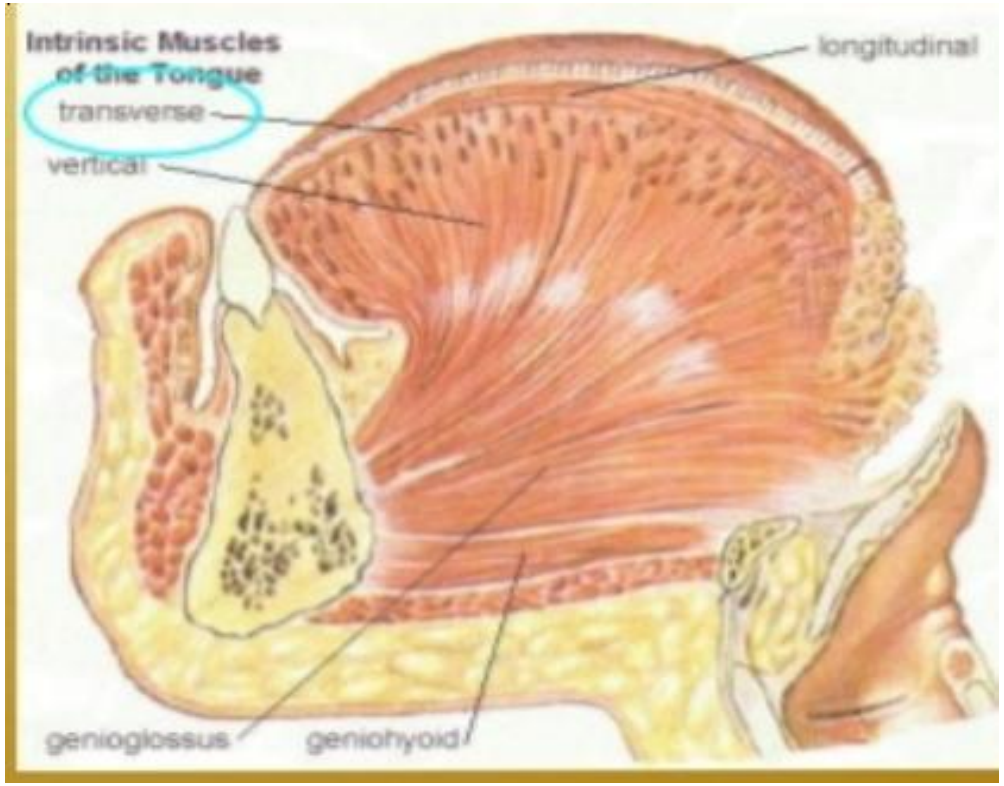
Origin - lingual septum

Insertion - lateral margin of tongue

Innervation - hypoglossal nerve (CN XII)

Blood supply - lingual branch of external carotid artery

Action - narrows and elongates tongue



Vertical muscle

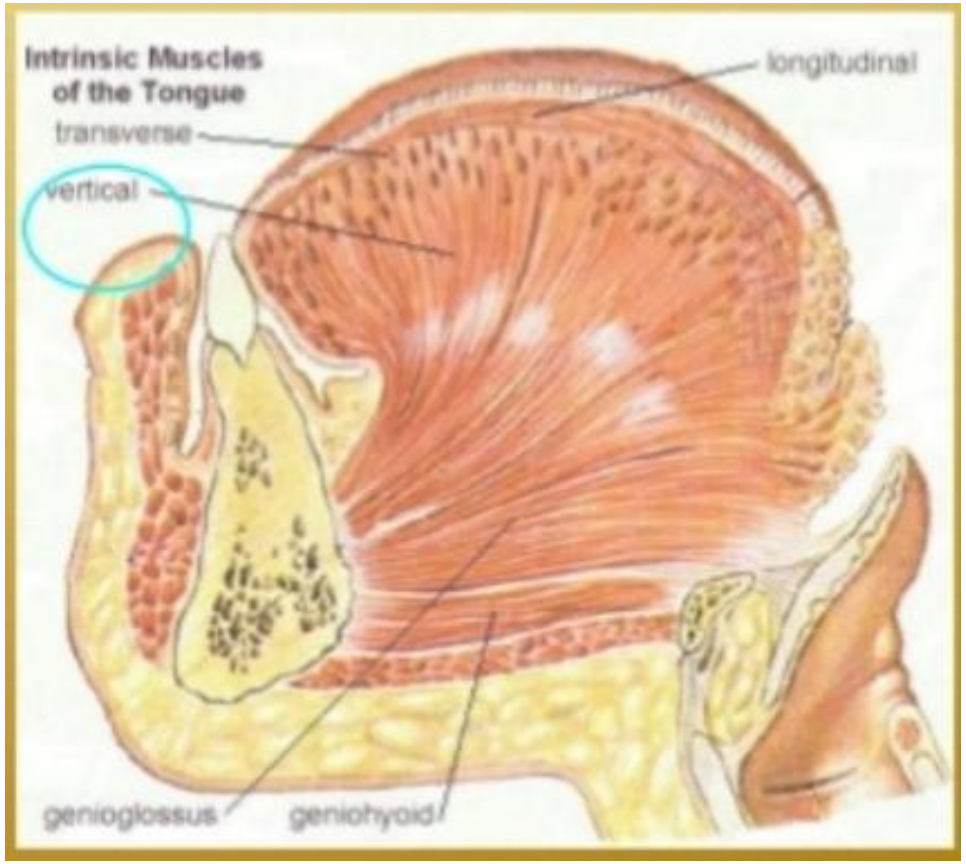
Origin - root of tongue, genioglossus muscle

Insertion - lingual aponeurosis

Innervation - hypoglossal nerve (CN XII)

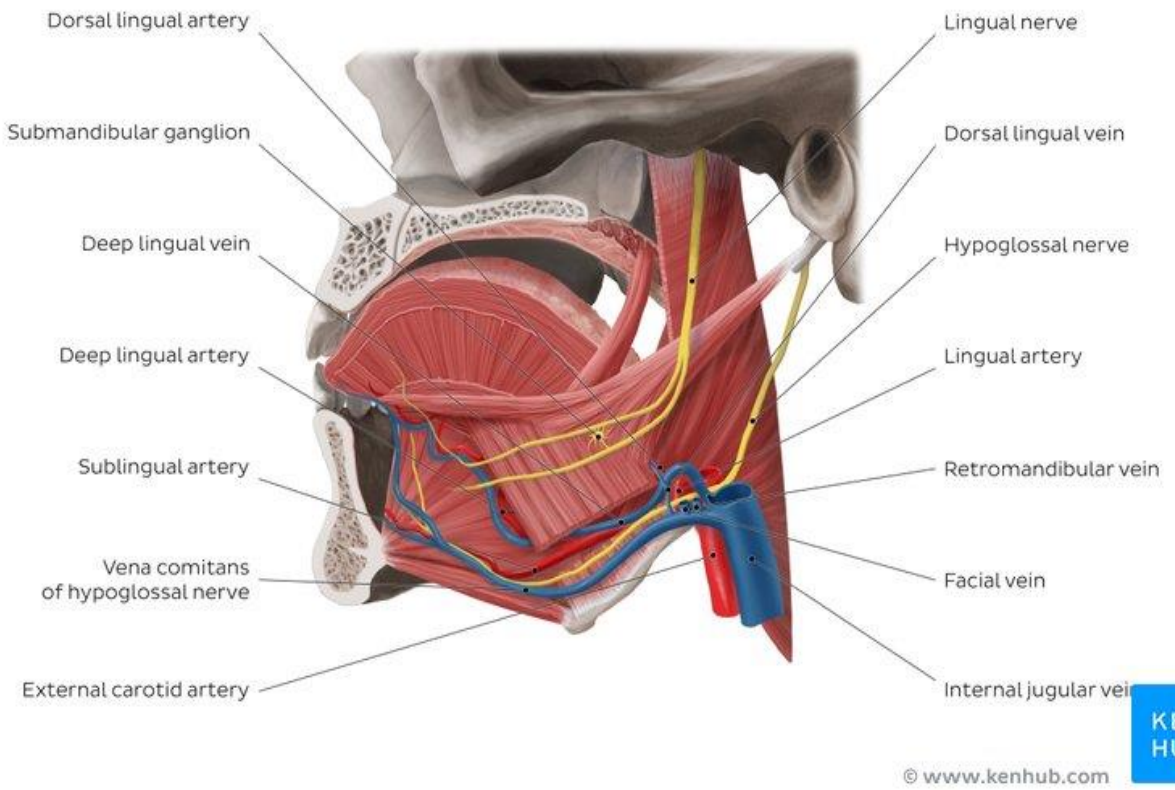
Blood supply - lingual branch of external carotid artery

Action - broadens and elongates tongue



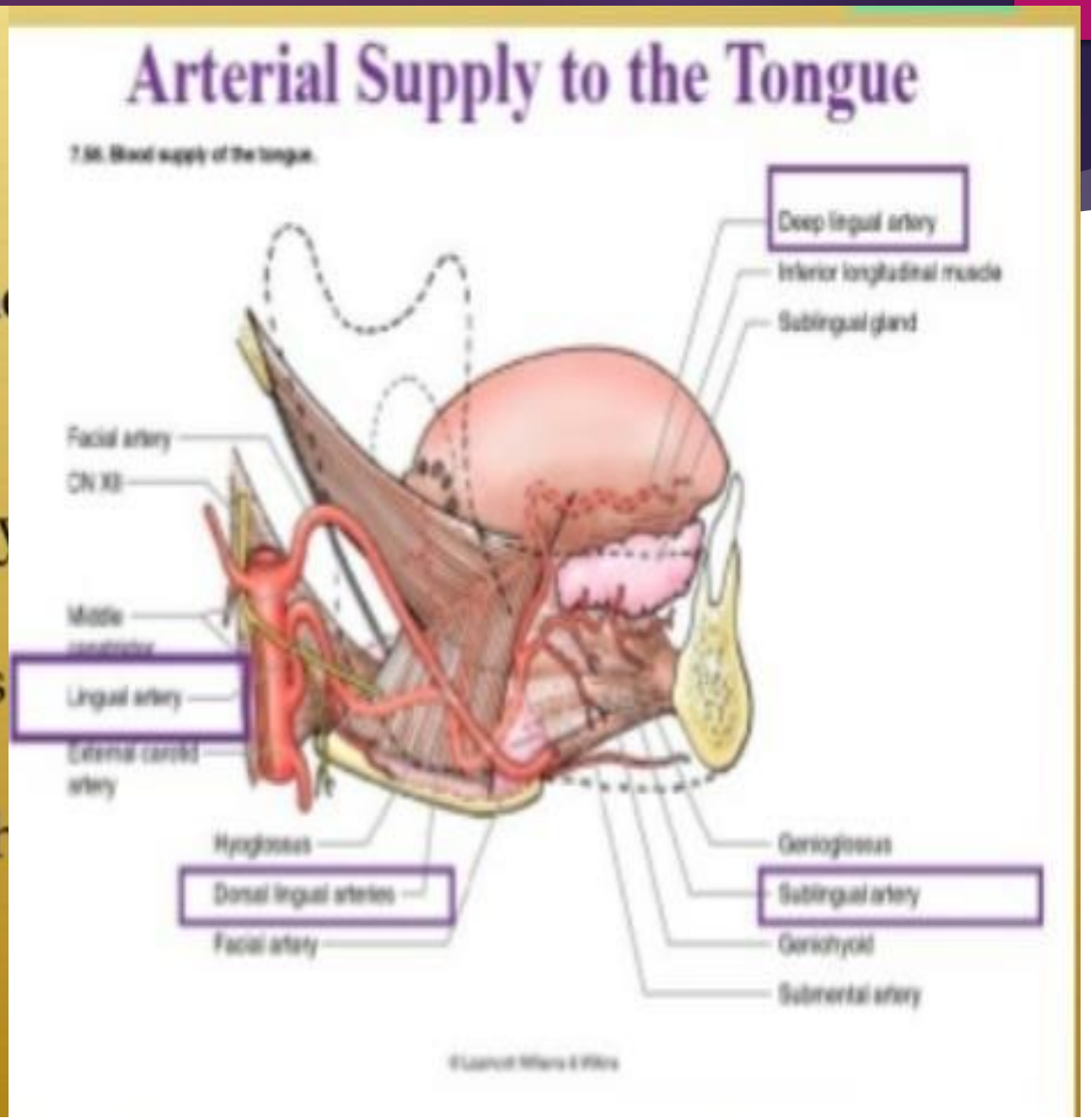
Blood supply

Lingual artery (dorsal lingual, sublingual, deep lingual arteries), ascending palatine, tonsillar, ascending pharyngeal arteries



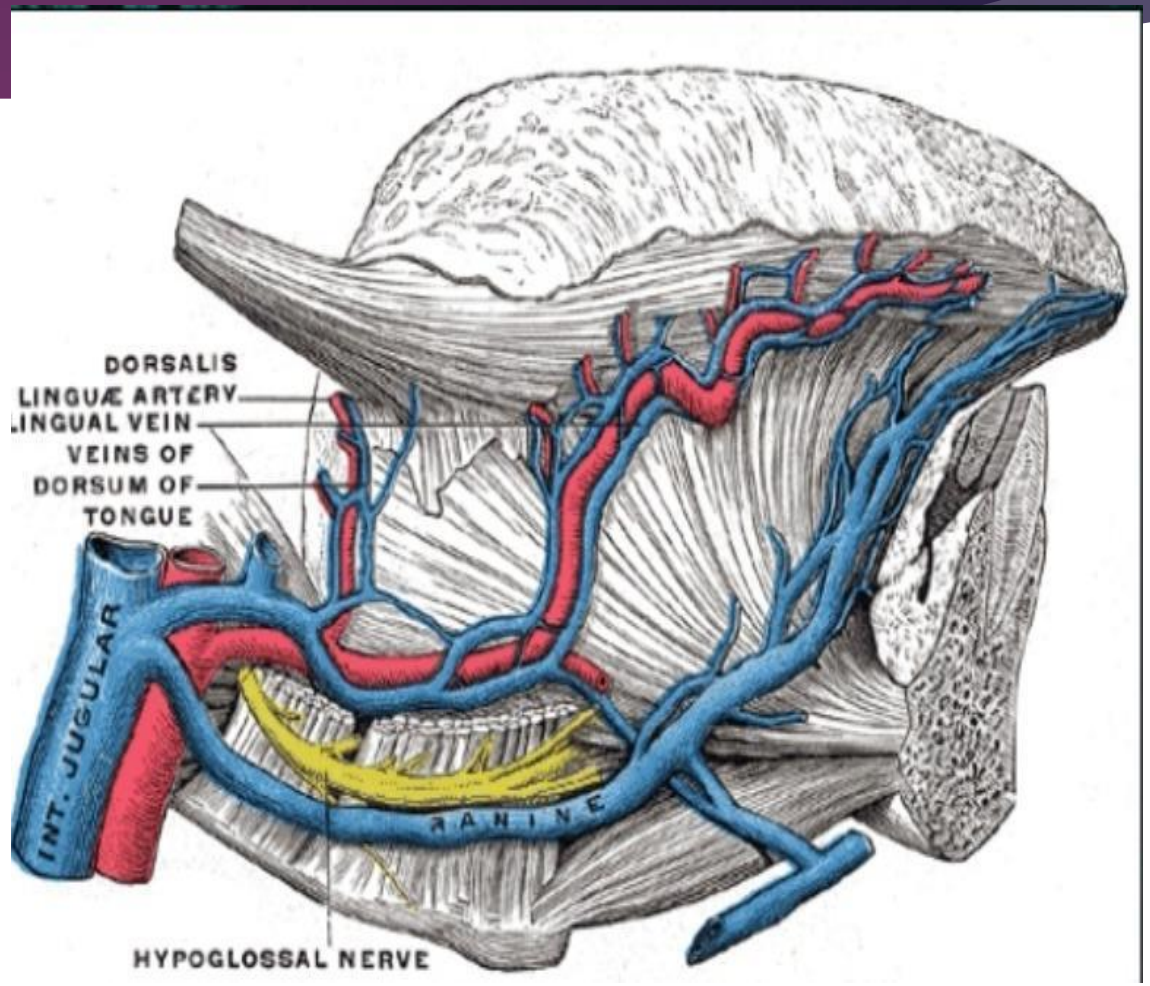
Lingual artery

- A branch of external carotid artery(after passing deep to the hyoglossus muscles)
- Divides into :
- Dorsal lingual arteries: supply posterior part
- Deep lingual artery : supplies the anterior part
- Sublingual artery : supplies the sublingual gland and floor of the mouth



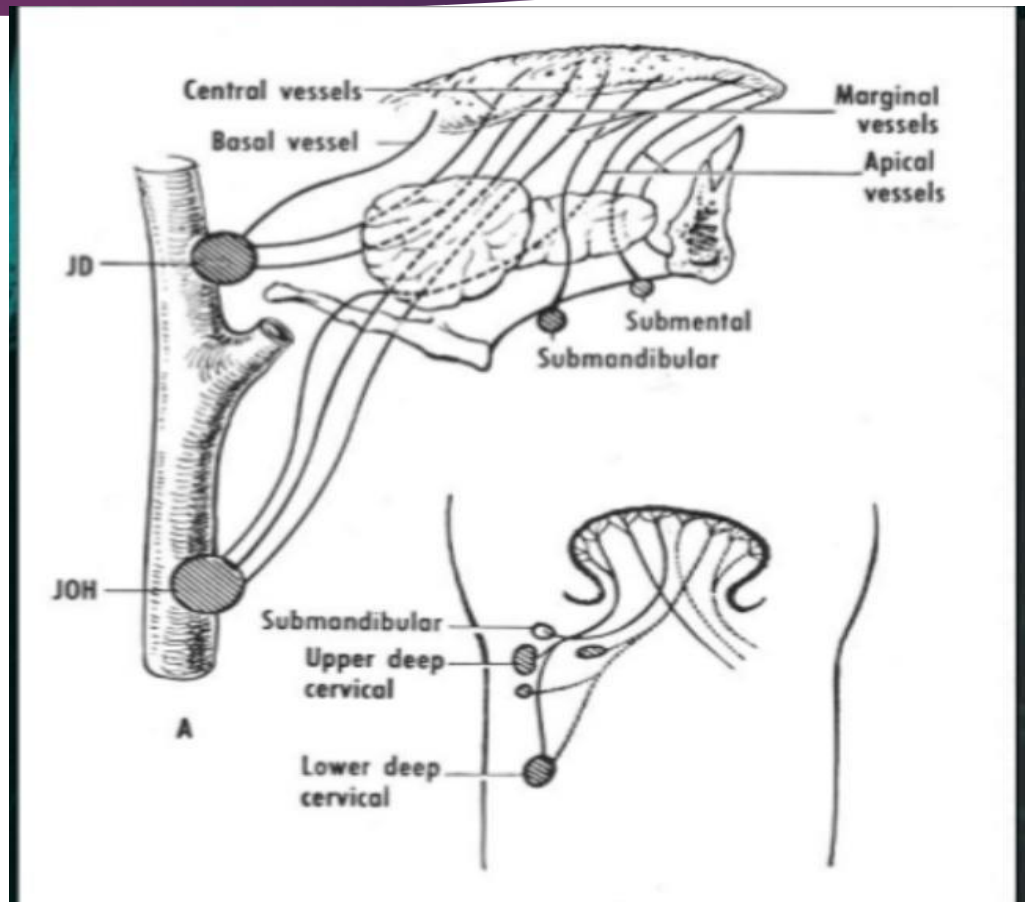
Venous drainage of tongue

- Dorsal lingual vein- drains the dorsum and sides of the tongue
- Deep lingual veins (Ranine veins) - drains the tip of the tongue and join sublingual veins from sublingual salivary gland
- All these veins terminate directly or indirectly into internal jugular veins

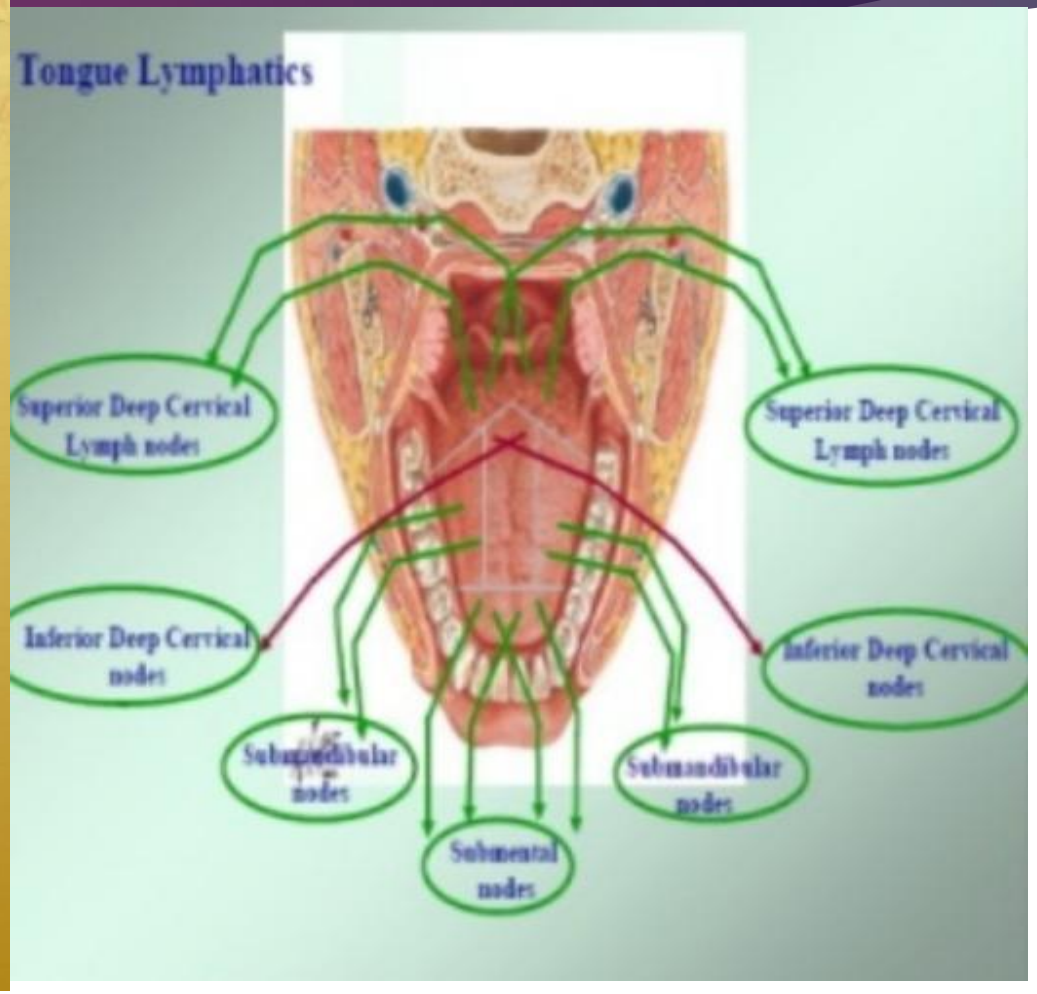


Lymphatic drainage

- Lymph from one side (esp. of the posterior side), may reach the nodes of the both sides of the neck (in contrast to the blood supply which remains **unilateral**)
- Tip - drain to submental nodes or directly to deep cervical nodes
- Marginal lymphatics from the anterior part tend to drain to ipsilateral submandibular nodes or directly to inferior deep cervical nodes



- Central lymphatics - drain to deep cervical nodes of either side
- Posterior part - drains directly and bilaterally to deep cervical nodes
- The deep cervical nodes usually involved: jugulodigastric and jugulo-omohyoid nodes
- All lymph from the tongue is believed to eventually drain through the jugulo-omohyoid node before reaching the thoracic duct or right lymphatic duct



Nerve supply of tongue

- Motor supply
- All muscles of tongue except palatoglossal are supplied by hypoglossal nerve
- Palatoglossus is supplied by pharyngeal plexus .

- Sensory supply
- Ant2/3 except vallare papilla

General sensation: lingual nerve

Special sensation : chorda tympani

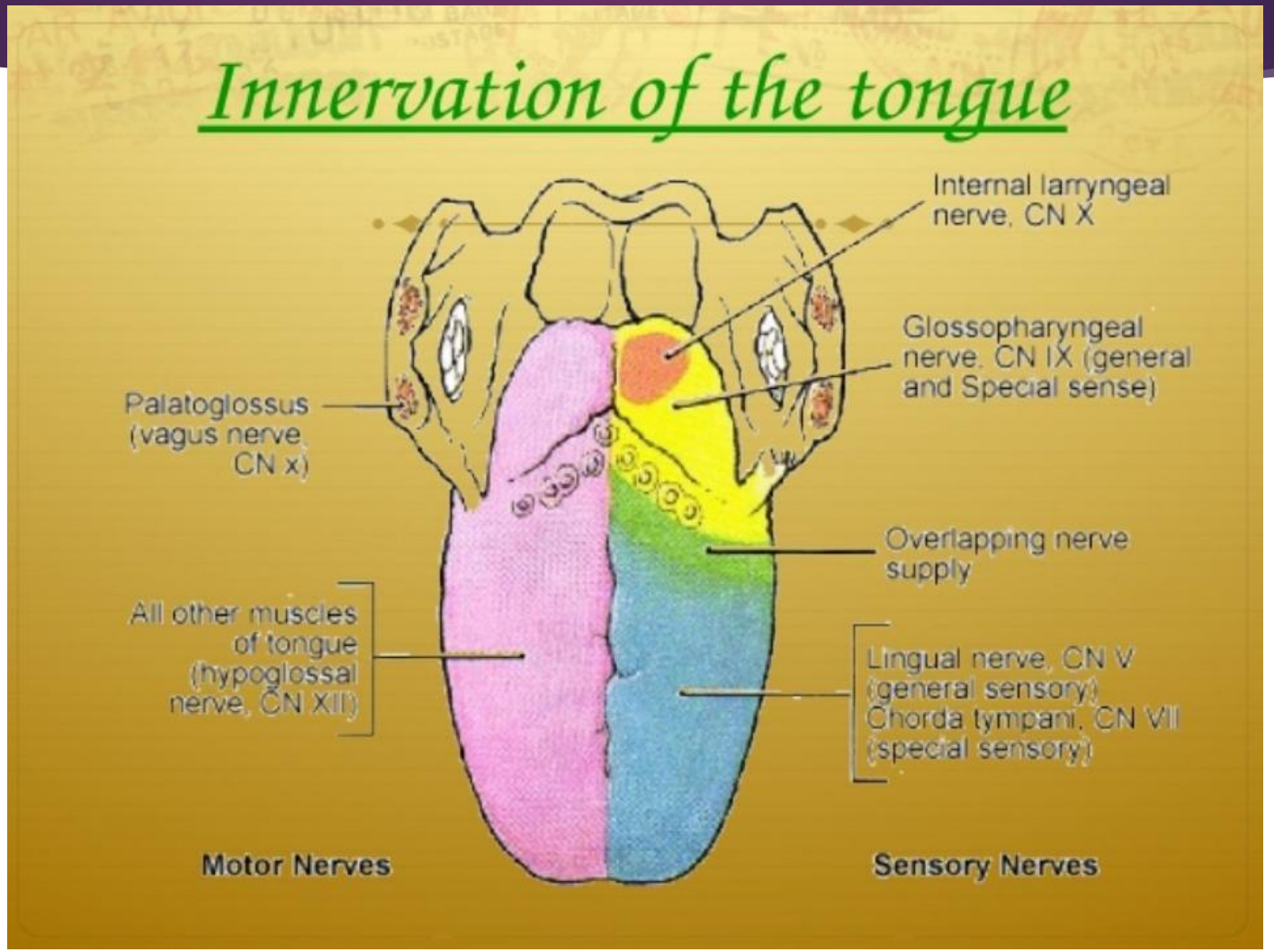
- Posterior 1/3 including vallare papilla

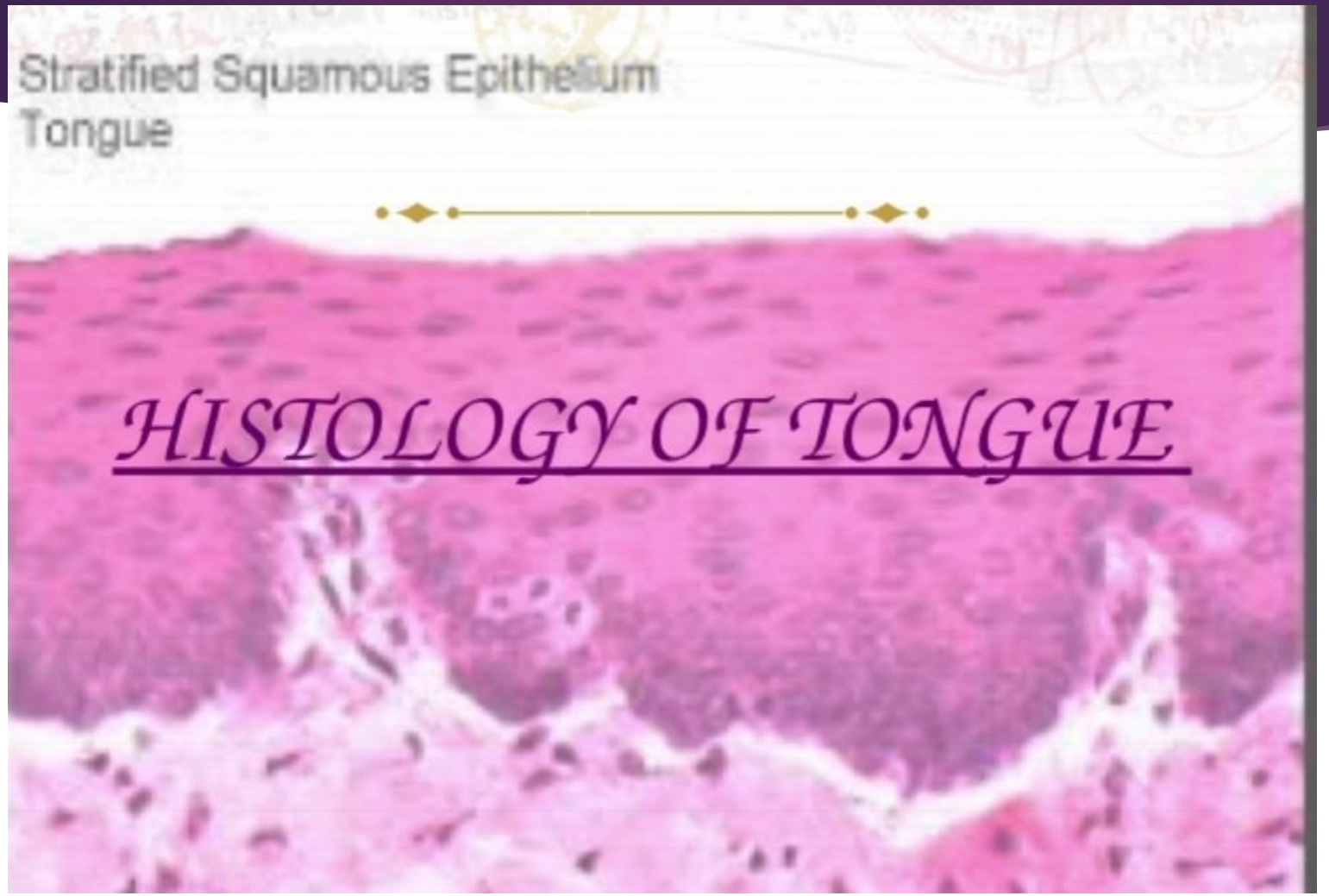
Both general and taste sensation by glossopharyngeal nerve

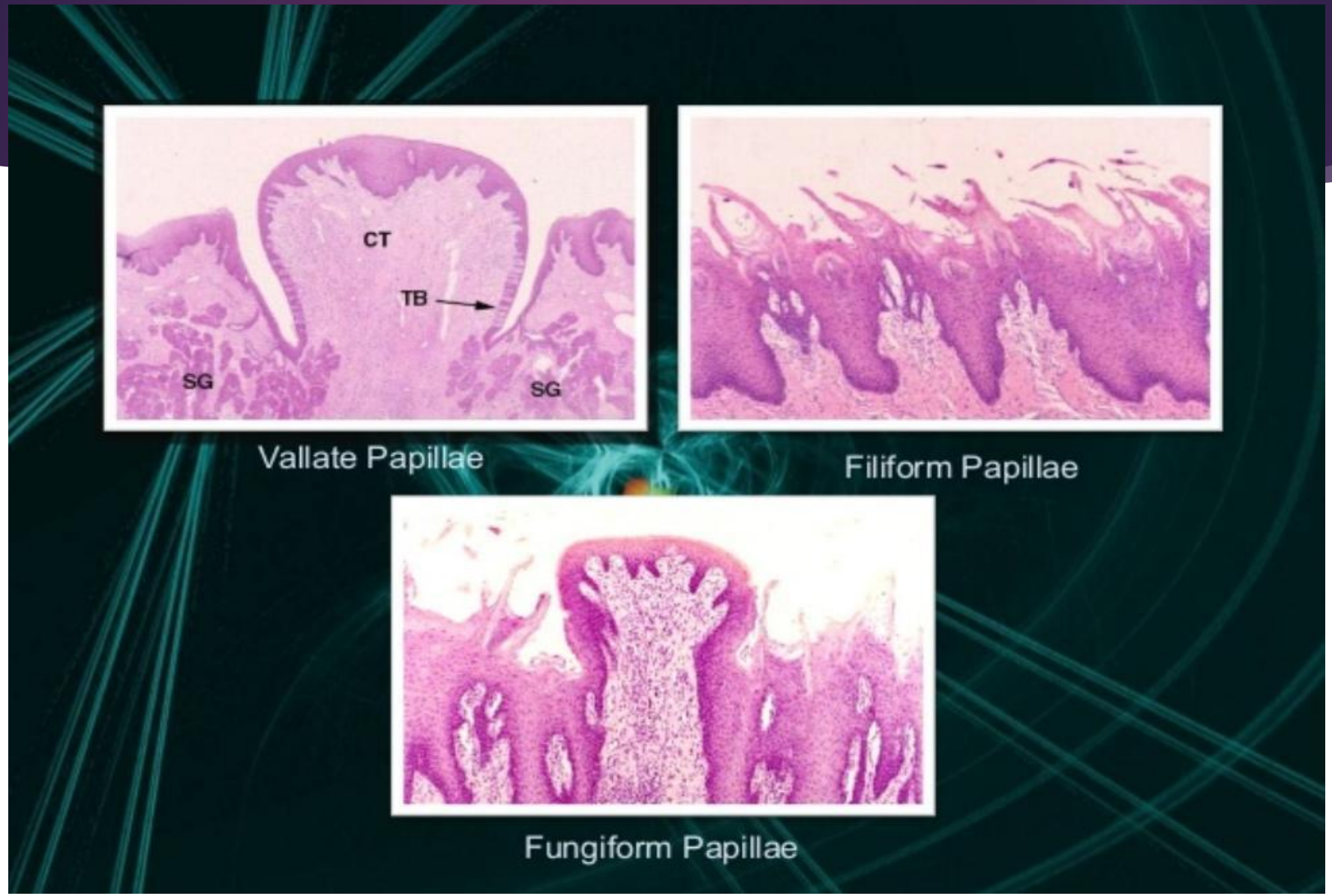
- Posterior most part

Internal laryngeal branch of vagus nerve

Innervation of the tongue




Stratified Squamous Epithelium
Tongue
HISTOLOGY OF TONGUE


CT
TB
SG
SG
Vallate Papillae
Filiform Papillae
Fungiform Papillae

Clinical examination of tongue

- Inspection
- The tongue is examined for:
  - Colour
  - Swelling
  - Ulcer
  - Coating
  - Size variation
  - Distribution of filiform and fungiform papilla
  - Crenations
  - Fissures
  - Atrophy or hypertrophy of papilla

- Frenal attachment and deviations as patient moves out the tongue



Palpation





Inherited, congenital, and developmental anomalies

- Variation in morphology
  - Ankyloglossia
  - Fissured tongue
  - Macroglossia
  - Hypoglossia
  - Lingual thyroid and thyroglossal duct

Partial ankyloglossia (tongue-tie)

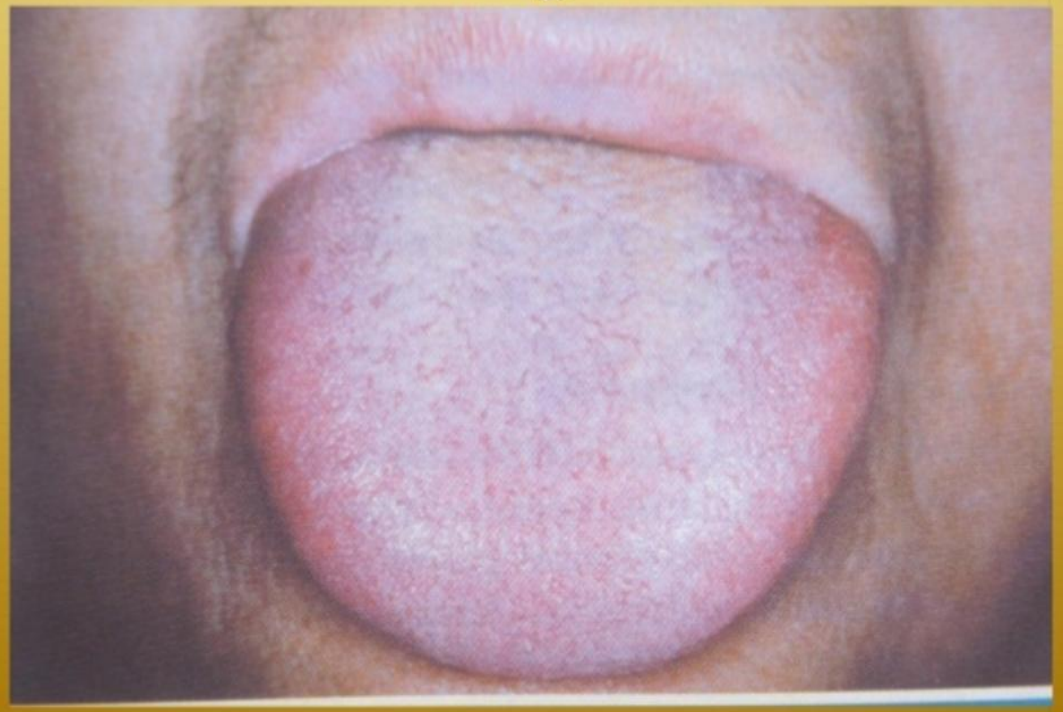


Microglossia (hypoglossia)

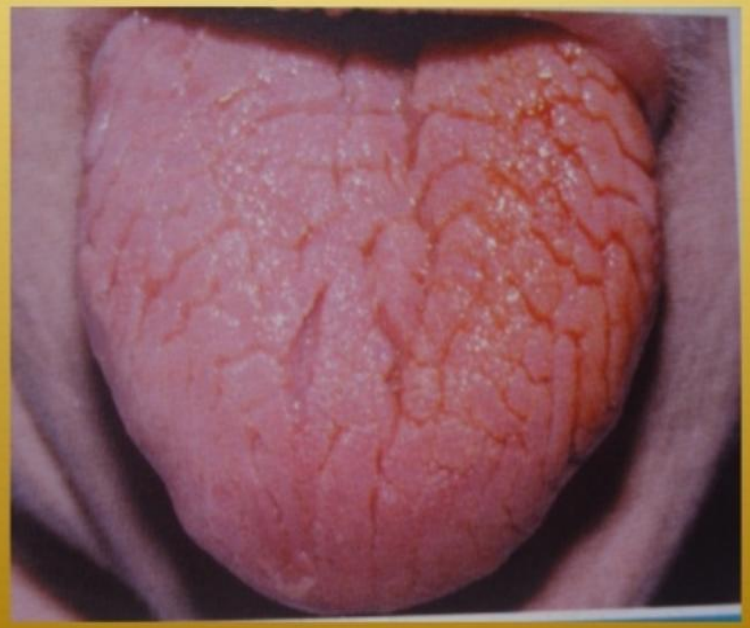
- ✓ Uncommon developmental condition of unknown origin characterized by abnormally small tongue
- ✓ Entire tongue may be missing (aglossia)
- ✓ length of the mandibular arch will be smaller due to the smaller size of the tongue.



Macroglossia



Fissured, plicated, or scrotal tongue



Lingual thyroid



Geographic tongue:

- Psoriasiform mucositis of the dorsum of the tongue
- Prevalence is 1% to 2%
- Irregular reddish areas of depapillation
- thinning of the dorsal tongue epithelium usually surrounded by a narrow zone of regenerating papillae -whiter than the surrounding tongue surface



Leukoplakia



Malignant tumors Of Tongue

